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Final Project Report

FeatherDB

Software Project Lab 1

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Submitted to

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Project Overview and Motivation

FeatherDB is a lightweight, file-based relational database engine written from scratch in **C++17**. It provides a transparent command-line interface (CLI) for interacting with data using SQL-like commands, specifically designed to demonstrate the internal mechanics of database systems without the overhead of massive, server-based RDBMS.

Motivation: The project was motivated by a desire to understand database internals through implementation rather than theoretical study alone. Foundational direction came from project-based learning resources, including open-source "Build Your Own X" repositories and related technical material. RBAC and ERD visualization were incorporated following recommendations from the project supervisor and panel judges during the proposal phase.

Problem Statement

The transition from a C programming course to a DBMS course presented an unexpected conceptual challenge. In C, program behavior is explicit, memory is managed manually, control flow is written out in full, and the relationship between code and execution is direct and traceable. SQL operates differently. A query such as `SELECT * FROM users WHERE age > 18` is declarative: it specifies what result is needed without describing how to obtain it. The mechanics of scanning a table, applying a filter, and returning a result set are handled internally, with no visible implementation for the programmer to examine.

This opacity became a source of genuine curiosity. Production database systems such as PostgreSQL and MySQL contain the answers somewhere within their codebases, but those systems span millions of lines of code developed over decades, making them impractical as learning resources. The result was a gap between using SQL as a tool and understanding it as a system.

FeatherDB was developed as an attempt to close that gap. It is not to produce a production-ready database, but to build one transparent enough that every stage of query processing could be examined, understood, and learned from.

Potential Solution

The central objective of FeatherDB was to construct a database engine in which every processing stage was implemented from scratch and could therefore be directly examined. The system follows a linear pipeline architecture: a SQL string is received, broken into tokens by the lexical analyzer, parsed into an Abstract Syntax Tree that encodes the statement's meaning, and passed to an executor that carries out the corresponding operations. Implementing this pipeline made the transformation from declarative SQL to concrete execution visible at every step.

Beyond parsing, the system was designed to behave as a functional relational database. FeatherDB enforces primary key uniqueness, unique column constraints, and foreign key referential integrity with a RESTRICT deletion policy. Implementing these constraints from scratch gave the team a concrete understanding of how relational integrity is maintained at the engine level rather than as an abstract guarantee.

Support for nested subqueries: both in **FROM** and **WHERE** clauses. These were included because recursive query structures represent one of the more non-trivial aspects of SQL execution. The **ORDER BY** clause is handled through an in-memory quicksort, a point where the team's prior experience with algorithmic thinking in C proved directly applicable.

The Role-Based Access Control system was incorporated following guidance from the project supervisor. Databases govern not only how data is stored but also who may access and modify it. FeatherDB implements secret-key authentication, role creation, and table-level permission assignment, with the access control layer sitting between the parser and the storage engine. The ERD viewer, recommended by one of the panel judges, provides terminal-based visualization of table relationships, automatically classifying foreign key connections as 1:1, 1:N, or N:N from schema metadata alone.

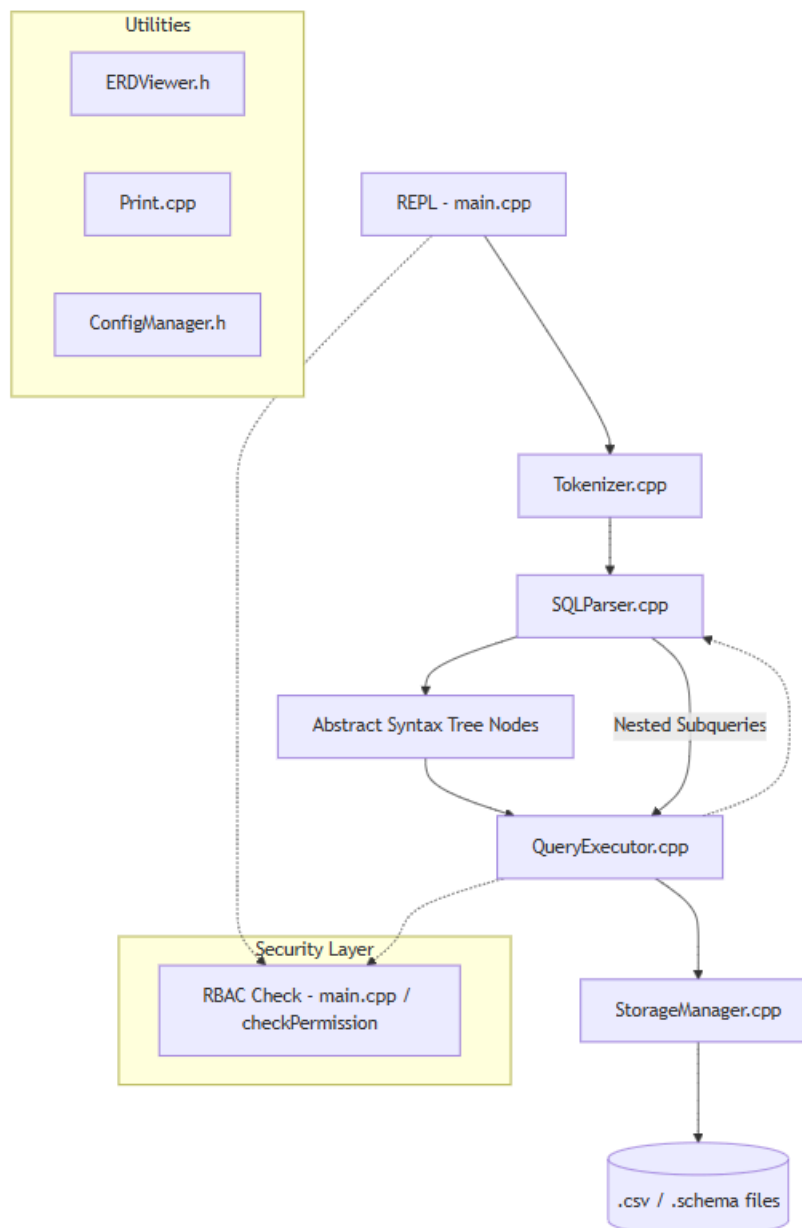
Data persistence is handled through a two-file format per table: CSV files for row data and schema files for structural metadata. This required the team to reason carefully about what state a database must preserve across sessions beyond the rows themselves.

Project Architecture and Flow

FeatherDB utilizes a decoupled, modular pipeline architecture. Every command flows through a strict verification and execution sequence:

REPL → Tokenizer → SQLParser → AST → QueryExecutor → StorageManager → CSV Files

- **REPL:** Captures user input and handles meta-commands (e.g., `.help`, `.tables`, `.erd`).
- **Parser & AST:** A recursive descent parser that validates syntax and builds a hierarchical tree representing the query.
- **Storage Manager:** Acts as the persistence layer, performing CRUD operations on physical `.csv` and `.schema` files in the `/db` directory.



Implemented Features

Core SQL Operations

- **CREATE TABLE** – Defines a new table schema with column types and constraint declarations. The schema is written to disk immediately and used to validate all subsequent operations on that table.
- **INSERT** – Adds a new row after running it through the full constraint pipeline – primary key uniqueness, unique column checks, and foreign key existence – before any write reaches disk.
- **SELECT** – Retrieves data with support for column projection, **WHERE** filtering, nested subqueries in both the **FROM** and **WHERE** clauses, and **ORDER BY** sorting.
- **UPDATE** – Locates rows matching a **WHERE** condition and modifies the specified columns, re-checking uniqueness and referential constraints on the new values before saving.
- **DELETE** – Removes matching rows, but refuses the operation entirely if any other table holds a foreign key reference to the target row (RESTRICT policy).
- **DROP TABLE** – Permanently removes both the data file and the schema file for a table from the **db/** directory.

Constraint Enforcement

- **PRIMARY KEY** – Enforces uniqueness and non-nullability on the designated column. Only one primary key per table is permitted.
- **UNIQUE** – Extends uniqueness guarantees to any column independently of the primary key. Multiple unique columns per table are supported.
- **FOREIGN KEY (REFERENCES)** – Links a column to the primary key of another table. A RESTRICT deletion policy is enforced – a parent row cannot be deleted while child rows still reference it.
- **Automatic Validation** – All three constraint types are checked on every DML operation before any write is committed to disk.

Advanced Query Features

- **Subqueries in FROM** – A nested `SELECT` in the `FROM` clause is treated as a derived table. The inner query is parsed and executed recursively before the outer query runs against its result.
- **Subqueries in WHERE** – A nested `SELECT` in the `WHERE` clause must resolve to a scalar value (one row, one column), which is then substituted back into the condition for evaluation.
- **Multi-level Nesting** – Both subquery forms support multiple levels of recursion, bounded by available call stack depth.
- **ORDER BY** – Results are sorted in-memory using a quicksort implementation applied to the result set before output. Ascending and descending order are both supported.
- **Comparison Operators** – Full support for `=`, `<`, `>`, `<=`, `>=`, and `!=` in `WHERE` clauses.

Role-Based Access Control (RBAC)

- **Authentication** – Roles are activated by entering a secret key at the REPL prompt. The active role is reflected in the prompt display (e.g., `featherDb> (admin)`).
- **Role Creation** – Admins can define new roles with custom secret keys using `CREATE ROLE ... WITH SECRET`.
- **Permission Granting** – Table-level privileges (`SELECT`, `INSERT`, `UPDATE`, `DELETE`, `CREATE`, `DROP`) can be assigned independently per role per table using `GRANT`.
- **Session Management** – The active role is tracked throughout the session and checked before every query reaches the execution stage.
- **Persistent Storage** – Role definitions and permission mappings are persisted to `roles.csv` and `permissions.csv` in the `db/` directory.

Relationship Visualization

- **ERD Generation** – The `.erd` meta-command analyzes all schema files and renders a text-based entity-relationship diagram in the terminal. Relationship types are inferred automatically from constraint metadata: a foreign key that is also `UNIQUE` or `PRIMARY KEY` is classified as 1:1; a plain foreign key as 1:N; a table where every column is a foreign key is treated as a junction table representing an N:N relationship.

Data Persistence

- **Two-File Storage Format** – Each table is stored as a `.csv` file for row data and a `.schema` file for column definitions and constraint metadata.
- **Immediate Persistence** – Every write operation is committed to disk without buffering. There is no staging layer or rollback mechanism.
- **Meta-Commands** – `.tables` lists all tables currently stored in the `db/` directory; `.schema <name>` reconstructs the `CREATE TABLE` statement from the schema file.

Tools & Technology

Programming Language

- **C++ (C++17)** – The engine is implemented entirely in C++. The choice was partly pragmatic; the team came from a C background and was already comfortable with explicit memory management and low-level data structure work, and partly intentional. Upon advice of the supervisor, we fixated on building a database in a language without a garbage collector, which meant that decisions around memory ownership and data layout had to be made consciously rather than delegated to a runtime.

Development Environment

- **Visual Studio Code** – Used as the primary editor throughout development. The C/C++ extension handled IntelliSense and debugging; the integrated terminal was used to run the compile script and test queries directly.

Build System

- **Custom Batch Scripts** – The project is compiled via `compile.bat`, which invokes G++ with the appropriate source files and flags. A separate `run_tests.bat` script runs the test case suite defined in `commands.txt`. No external build system (CMake, Make) was used. The scripts are short enough to read and understand in under a minute.

Version Control

- **Git / GitHub** – Git was used for version control across the three-member team. The repository is hosted on GitHub and still serves as the central point for coordinating contributions, tracking changes, and maintaining a recoverable project history.

Standard Library

- **C++ STL** – The implementation relies almost entirely on the standard library. `std::vector`, `std::map`, and `std::set` handle row collections, column metadata, and permission mappings. `std::string` is used throughout for token manipulation and value representation. File I/O uses standard `std::ifstream` and `std::ofstream` streams. No third-party libraries were introduced. An external dependency was only worth considering if the functionality genuinely couldn't be built from scratch, which in practice barely came up.

Diagrams & Presentation

- **Mermaid** – Used to produce architecture and data-flow diagrams. Diagrams are written as text and rendered from within the documentation, keeping them version-controllable alongside the code.
- **Canva** – Used for preparing project presentations.

Application Domain

Potential Use Cases

- **Educational Tool for Database Internals** – The primary use case FeatherDB was built for. Rather than treating SQL as a black box, the codebase exposes every stage of query processing, from tokenization through to disk writes (or files in our case), in a form that can actually be read and understood. It is particularly suited to students who have some programming background but have never looked inside a database engine before.
- **Teaching Relational Concepts** – FeatherDB can be used in classroom settings to demonstrate constraint enforcement, referential integrity, and relational algebra in a working system rather than through abstract examples. The ERD viewer makes schema relationships visible without requiring any external tooling.
- **Lightweight Personal Projects** – For small-scale applications that need basic SQL functionality without the overhead of setting up a full database server, FeatherDB's file-based storage and simple REPL interface are practical enough. It is not suitable for anything beyond low-volume, single-user use, but for prototyping or local tooling, it works fine.
- **Demonstrating RBAC Patterns** – The multi-role authentication and permission system can serve as a reference implementation for understanding how access control is layered into a database engine, which is a concept that often gets glossed over in introductory courses.

Target Audience

- CS students taking database or systems courses who want to see how SQL works under the hood
- Educators looking for a readable, self-contained codebase to walk through in lectures
- Developers who are curious about DBMS internals and want something small enough to explore in a weekend

Non-Target Applications

- Production systems of any kind, FeatherDB has no concurrency control, no transaction support, and no crash recovery
- Any dataset that does not fit comfortably in RAM
- Applications requiring ACID compliance, distributed operation, or high-throughput query processing

Project Timeline

The project spans fourteen weeks in total. The work plan is structured to ensure steady progress from foundational design to full implementation, testing, and presentation. Each phase contributes directly to the core objective of constructing a minimalist, C++-based SQL-like database engine with custom parsing, storage, and execution components.



In the final stretch, we enhanced system security by implementing a passkey-protected Role-Based Access Control (RBAC) system for managing administrative and user permissions. This was followed by a rigorous validation phase where we refined the parser to accurately handle multi-word arguments and enforce semicolon termination. Additionally, we expanded the engine's logical depth by enabling the **WHERE** clause to process essential mathematical comparisons such as `=`, `<`, `>`, `>=`, and `<=`.

Weeks 1-2: Planning and Team Formation

The first two weeks were dedicated to establishing the project's direction through brainstorming sessions, reviewing expectations, discussing feasibility, and forming the project team. During this phase, the team finalized the goal of building a lightweight database engine with minimal external library usage and clarified individual responsibilities.

Weeks 3–4: Proposal Development and Initial Setup

These weeks focused on preparing the project proposal and presenting it to the instructors. The feedback received during this stage guided refinements to the project scope, vocabulary used in parser design, and the plan to transition from Python to C++.

In parallel, the team began setting up the initial project structure in C++, organizing modules, selecting coding conventions, and preparing the environment for test-driven development.

Weeks 5–6: Core Fundamentals and Foundational Components

This stage is dedicated to laying the technical foundation of the system. The team will implement the core utilities required for a minimal database engine, including:

1. Custom string handling utilities (if required by the final design),
2. Lexing/tokenization logic,
3. Preliminary interfaces for parser modules,
4. Initial abstractions following object-oriented principles (e.g., Token, Node, StorageManager interfaces).

The goal of this phase is to ensure that subsequent layers (storage and parsing) have a stable, well-defined base.

Weeks 7–8: Storage Engine and Parser Construction

The storage subsystem will be developed during this period, focusing on file-based persistence, record formatting, and basic table operations.

Alongside the storage layer, the team will work on the parser—constructing the Abstract Syntax Tree (AST) with at least three levels of nesting, refining syntactic rules, and establishing the link between parsed queries and upcoming execution logic.

A ***mid-project progress presentation*** is planned between Weeks 8–9 to evaluate alignment with expectations and receive additional feedback.

Weeks 9–10: Command-Line Interface (CLI) Development

These weeks are devoted to building the interactive command-line interface that users will interact with.

Tasks include:

- Designing input handling and communication with the parser,
- Providing meaningful error messages,
- Integrating the parser with the storage and execution pipeline,
- Ensuring smooth round-trip execution of simple queries.

This phase turns the internal modules into a usable tool.

Weeks 11–12: Extended Features and Role-Based Access Control

After the MVP becomes functional, focus shifts toward the additional features recommended during the proposal feedback. These include:

- designing a role-based access control (RBAC) system, such as admin/user roles with permission boundaries;
- supporting more complex SQL-like operations (e.g., nested queries aligned with the three-level AST requirement);
- strengthening internal modularity and improving parser vocabulary and error handling.

These enhancements expand both the realism and academic value of the project.

Weeks 13–14: Testing, Documentation, and Final Presentation

The final two weeks concentrate on rigorous testing following the test-driven development approach. Unit tests, integration tests, and behavior tests will verify correctness across all components—lexing, parsing, storage, CLI, and RBAC.

Simultaneously, the team will prepare project documentation, including design explanations, module descriptions, usage guidelines, and analysis of limitations.

The work concludes with the final project presentation, demonstrating the database engine's features, architecture, and evolution from proposal to implementation.

Action Based on Feedback (Progress & Evolution)

During the proposal presentation, feedback from the course supervisors and judges led to several concrete changes to the project scope and implementation:

- **Transition to C++** – The original proposal planned to use Python, primarily because it would have been quicker to get something working. Following feedback from the judges, the implementation was shifted to C++. This turned out to be the right call since working closer to the hardware made the internal mechanics of the database more visible, which aligned better with what the project was trying to demonstrate.
- **RBAC Integration** – One of the judges suggested adding an access control layer to make the project more complete as a database system. This led to the implementation of the secret-key authentication system and the role-permission model, which became one of the more interesting parts of the project to build.
- **ERD Visualization** – A judge recommended including a way to visualize table relationships, which resulted in the `.erd` utility. Implementing the relationship detection logic and distinguishing 1:1, 1:N, and N:N from constraint metadata alone ended up being a worthwhile addition.
- **Mathematical Foundations** – The project supervisor advised incorporating a formal treatment of the underlying concepts. This shaped how the project addresses the theoretical basis of the system, grounding the implementation in relational algebra and constraint theory rather than treating it as purely a coding exercise.

Contribution

Collaborative Achievements

All three members participated in design discussions throughout the project, collectively deciding on the system architecture, constraint enforcement strategy, and RBAC structure. Internal documentation, the project README, and the report were developed collaboratively. The build system, coding conventions, and REPL integration were finalized as a group, as were code reviews conducted in preparation for the presentation. QueryExecution and Debugging were also collaborative efforts.

Individual Roles

Md. Saif-AI Sarker (230042106): Testing & Query Execution

Contributed to the query execution layer and was responsible for the project's testing infrastructure. This included unit tests for the Tokenizer, Parser, and AST validation, integration tests for the QueryExecutor covering constraint scenarios, and storage persistence tests verifying CSV read/write correctness. Test automation was set up via `run_tests.bat`, and reproducible bug cases were documented for debugging purposes. Also contributed to the implementation of core DML operations within the QueryExecutor.

M Safwan Hasan Khan (230042117): Storage Engine & Utilities

Contributed to the persistent storage layer and the supporting utilities. This covers the StorageManager (`StorageManager.cpp/h`), which handles CSV serialization and deserialization, schema file management, and data synchronization. Supporting utilities include `ConfigManager.h` for environment-based configuration, `Print.cpp/h` for formatted query output, and the ERD Viewer (`ERDViewer.h`), which automatically detects and visualizes 1:1, 1:N, and N:N relationships from schema metadata.

Wasi Omar Syed (230042143): SQL Parser & Query Execution

Contributed to the SQL processing pipeline, including the Tokenizer (`Tokenizer.cpp/h`), which performs lexical analysis and classifies SQL input into token streams, and the SQLParser (`SQLParser.cpp/h`), a recursive descent parser that constructs Abstract Syntax Trees from tokenized input and handles nested subqueries in both `FROM` and `WHERE` clauses. Also contributed to the QueryExecutor alongside the rest of the team, particularly around expression evaluation and ORDER BY sorting.

Future Work

The current implementation covers the core of what a relational database needs to do, but there is plenty of room to grow. The most natural next step would be adding **JOIN** support. Right now multi-table queries require subquery workarounds, and a proper **INNER JOIN** syntax would make FeatherDB significantly more expressive. Closely related is aggregate function support (**COUNT**, **SUM**, **AVG**, **MIN**, **MAX** with **GROUP BY**), which is probably the most commonly missed feature when working with the current system.

On the SQL completeness side, transaction support with **ROLLBACK** would be a meaningful addition. As of now, every operation commits immediately, which means there is no way to undo a sequence of changes if something goes wrong midway. Views, triggers, and stored procedures, etc., would push FeatherDB further toward a complete SQL environment, though those are substantially more complex to implement cleanly.

The biggest architectural limitation is performance. Every query currently does a full table scan with no indexing, and every operation reloads the table from disk. Introducing a B-tree index structure and a basic buffer pool would address both of these, and would also make the codebase more interesting from a learning perspective; those are concepts that come up in every database course but rarely get implemented from scratch.

Further out, multi-threading and concurrent session support would require rethinking the storage layer significantly, but would make FeatherDB viable for slightly more realistic use cases beyond single-user exploration.

Conclusion

FeatherDB was developed in response to a straightforward question of what actually occurs when a SQL query is executed. The project's outcome is a functional relational database engine which is comprising of a tokenizer, parser, Abstract Syntax Tree, query executor, constraint enforcement layer, access control system, and ERD visualizer; all implemented from scratch in C++.

As the team's first substantial software project, undertaken in the third semester, the development process involved navigating unfamiliar design decisions, revising earlier implementation choices, and resolving non-trivial bugs. The experience of constructing a working system, rather than studying one, provided a depth of understanding that theoretical coursework alone does not readily offer. External resources including technical content on YouTube, open-source reference projects, and AI-assisted development tools were consulted throughout, not as substitutes for understanding but as aids in working through design tradeoffs and implementation challenges.

FeatherDB is not intended as a production system, nor does it claim to be. It is a transparent, readable implementation of core database principles. Measured against what the project set out to achieve, that is precisely what it needed to be.

Source Repository

GitHub Link: <https://github.com/SaifSarker11/FeatherDB>